EFFICACY AND SAFETY OF PUSH-DOSE NOREPINEPHRINE FOR INTRAOPERATIVE HYPOTENSION MANAGEMENT IN NON-OBSTETRIC SURGERY: A SYSTEMATIC REVIEW

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◆ BACKGROUND

Intraoperative hypotension (IOH), defined as MAP below 65 mmHg, occurs in up to 30% of non-cardiac surgeries and is independently associated with myocardial injury, acute kidney injury, and increased mortality. Push-dose (bolus) norepinephrine (NE) has emerged as a practical vasopressor option; however, evidence supporting its efficacy and safety remains limited.

◆ OBJECTIVE

To systematically evaluate the efficacy and safety of push-dose norepinephrine for IOH management in non-obstetric surgery compared to other vasopressors.

◆ METHODS

A systematic review following PRISMA 2020 was conducted. PubMed, Scopus, and ProQuest were searched. Inclusion criteria: adults (≥18 years), non-obstetric surgery, RCT design. Quality assessment used the Cochrane RoB 2 tool.

Search: "Norepinephrine" AND ("push dose" OR "bolus") AND ("intraoperative hypotension" OR "surgery")

Records: 643 identified → 363 after deduplication → 5 full-texts → **3 RCTs included**

◆ RESULTS

Three RCTs (n=388) were included. Push-dose NE 10 mcg/bolus (10 mcg/mL, peripheral IV) was consistent across all studies.

- **vs. Ephedrine:** NE bolus significantly improved MAP (98.9±12.7 vs 91.5±10.9 mmHg; **p<0.001**) and reduced hypotension episodes (**p=0.005**).
- **vs. NE Continuous Infusion:** Bolus NE showed comparable hypotension burden in low-to-moderate risk patients (p=0.070) with 66% lower cumulative drug dose (**p<0.001**). Continuous infusion provided better haemodynamic stability in high-risk patients (ARV-MAP; **p<0.001**, d=1.00).
- **Safety:** No significant overshoot hypertension or arrhythmia differences across groups. Peripheral IV administration was safe with zero extravasation complications.
- **Certainty of Evidence:** GRADE Low (⊕⊕○○). All studies: RoB 2 "Some Concerns".

Study	Population	Intervention vs Comparator	Key Findings (MAP / Safety)
Hassani (2018)	Hypertensive, GA Spinal Surgery	NE 10 mcg vs Ephedrine 5 mg	NE superior MAP (**p<0.001**), fewer doses
Thomsen (2026)	ASA ≥2, Low–Mod Risk	NE Bolus vs Continuous Infusion	AUC MAP <65 setara (p=0.07), 66% less dose
Vokuhl (2025)	ASA II–IV, High Risk	NE Bolus vs Continuous Infusion	Infusion better stability (**p<0.001**), safe IV

⊕⊕○○ MAP reversal · GRADE Low

⊕⊕○○ HTN overshoot · GRADE Low

⊕○○○ Time to MAP target · GRADE Very Low

— Arrhythmia · No evidence

◆ CONCLUSIONS

Push-dose NE 10 mcg via peripheral IV is **effective and safe** for IOH management in non-obstetric surgery. It is **superior to ephedrine** and offers comparable efficacy to continuous infusion in low-to-moderate risk patients. For **high-risk patients** with continuous arterial monitoring, continuous infusion may provide better haemodynamic stability. Multicenter RCTs with standardised outcome definitions are needed.

REFERENCES

1. Sessler DI, et al. *Anesth Analg.* 2019;128(4):717–727.
2. Hassani V, et al. *Anesth Pain Med.* 2018;8(4):e79626.
3. Thomsen KK, et al. *Br J Anaesth.* 2026;136(4):1137–1144.
4. Vokuhl C, et al. *Br J Anaesth.* 2025;135(4):878–885.
5. Berkenbush M, et al. *West J Emerg Med.* 2024;25(5):708–714.
6. Page MJ, et al. *BMJ.* 2021;372:n71.
7. Sterne JAC, et al. *BMJ.* 2019;366:l4898.